DSP · MOCK

DSP mock A

Practice questions · Day 22 · Sat 14 Nov · pack DX1

Instructions. Time allowed: **2 hours**. Open book: hard copy material is permitted. Attempt all questions. There are **60 marks** in total. Marks for each part are given in brackets. Show your working: method marks are given even when a number is wrong.

Budget about **2 minutes per mark**. Leave 5 minutes at the end to check units, ROCs and signs. If you get stuck on an early part, use the "assuming the answer to part (...) is ..." value and keep going.

This is a mock sat 4 days before the DSP exam. It covers L02–L10: about 25% on L02–L04 (Q1–Q2, 15 marks) and 75% on L05–L10 (Q3–Q8, 45 marks). Notation follows the course: F and Ω for continuous time, f and ω for discrete time, and $\operatorname{sinc} x = \sin x / x$.

Do not scroll past the horizontal rule until time is up.

Question	Topic	Lectures	Marks
1	Sampling, aliasing and system properties	L02–L03	8
2	LCCDE, impulse response and convolution	L04	7
3	Frequency response and DTFT properties	L05	7
4	DFT, circular convolution and resolution	L06	8
5	z-transform, ROC and inverse	L07–L08	8
6	Notch and comb filters	L09	7
7	Butterworth IIR design by bilinear transform	L10	9
8	Windowed linear-phase FIR design	L10	6
	Total		**60**

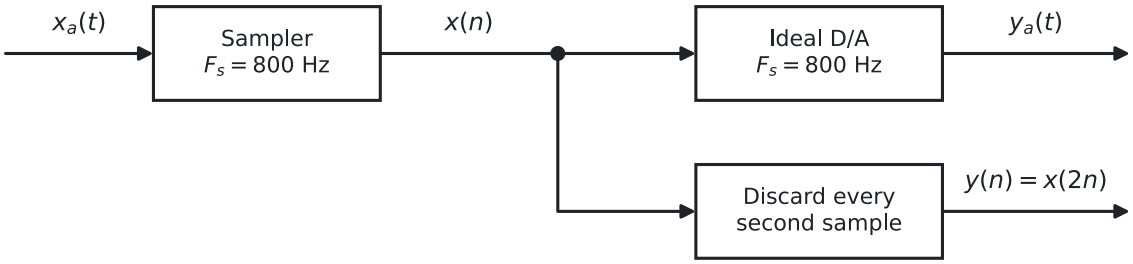
Question 1 [8 marks]

Sampling and system properties

The analogue signal

$$x_a(t) = 3\cos(300\pi t) + 2\sin(1100\pi t) \qquad (t \text{ in seconds})$$

is sampled at $F_s = 800$ Hz to give $x(n) = x_a(nT)$.



Question 1: the sampling chain, with the ideal D/A of part (c) and the sample-dropping system of part (d)

a. Determine the Nyquist rate for $x_a(t)$. Is $F_s = 800$ Hz adequate? **[1]**

b. Write $x(n)$ using only digital frequencies in the principal range $-\pi \le \omega \le \pi$. **[2]**

c. If $x(n)$ is passed through an ideal D/A converter at $F_s = 800$ Hz, what signal $y_a(t)$ is reconstructed? **[1]**

d. Every second sample of $x(n)$ is discarded, giving $y(n) = x(2n)$. Show that $y(n)$ is a single sinusoid. Write it as $A\cos(\omega_0 n + \phi)$ and give A, ω_0 and ϕ. **[2]**

e. For each system, state whether it is linear, time invariant, causal, static (memoryless) and BIBO stable. Give a one-line reason for each "no". **[2]**

f. $y(n) = x(n) - 0.5\,n\,x(n-1)$

ii. $y(n) = |x(n+1)|$

Question 2 [7 marks]

DT LTI systems

A causal DT LTI system is described by the LCCDE

$$y(n) - \tfrac{1}{4}y(n-1) - \tfrac{1}{8}y(n-2) = x(n) + x(n-1).$$

a. Find the characteristic modes of this system. Hence state whether it is BIBO stable. **[1]**

b. Using recursion, evaluate $y(n)$ for $0 \le n \le 3$ when the input is $x(n) = \left(\tfrac{1}{3}\right)^n u(n)$. Assume $y(n) = 0$ for $n < 0$. **[2]**

c. Find the impulse response $h(n)$. **[2]**

d. Assuming the answer to part (c) is $h(n) = \left[2\left(\tfrac{1}{2}\right)^n - \left(-\tfrac{1}{4}\right)^n\right]u(n)$, evaluate the convolution $y(n) = h(n) * x(n)$ in closed form, where $x(n) = \left(\tfrac{1}{3}\right)^n u(n)$. Check your answer against part (b) at $n = 0$ and $n = 1$. **[2]**

Question 3 [7 marks]

Frequency response

An FIR system is described by

$$y(n) = 0.5\,x(n) + x(n-1) + 0.5\,x(n-2).$$

a. Find the frequency response $H(\omega)$. Write it as a real amplitude times a linear-phase term, and give $|H(\omega)|$ and $\theta(\omega)$. **[2]**

b. Does this system have linear phase? State its group delay. **[1]**

c. Without computing any integral explicitly, evaluate $\displaystyle\int_{-\pi}^{\pi} |H(\omega)|^2\, d\omega$. **[1]**

d. Determine the steady-state output for the input **[3]**

$$x(n) = 4 + 2\cos\!\left(\tfrac{\pi}{2}n + \tfrac{\pi}{6}\right) + 3\cos\!\left(\tfrac{2\pi}{3}n\right) + 5(-1)^n.$$

Question 4 [8 marks]

The DFT

Consider the sequences

$$x_1(n) = \{\underset{\uparrow}{1}, 2, 0, -1\}, \qquad x_2(n) = \{\underset{\uparrow}{2}, 1, 1\}.$$

a. Compute the linear convolution $x_1(n) * x_2(n)$. **[1]**

b. Using your result from part (a), find the 4-point circular convolution $x_1(n) \circledast_4 x_2(n)$. What is the smallest DFT length N for which circular convolution gives the linear convolution? **[2]**

c. Compute the 4-point DFT $X_1(k)$ of $x_1(n)$. **[2]**

d. A signal contains two tones at 1.00 kHz and 1.03 kHz. It is sampled at $F_s = 8$ kHz and analysed with an $N = 2^m$-point DFT. Find the minimum N that gives a frequency resolution of 30 Hz or better, and the corresponding minimum record length T_0. **[2]**

e. A colleague records only $L = 200$ samples and zero-pads them to $N = 512$ before taking the DFT. Will the two tones be resolved? Explain in one or two sentences. **[1]**

Question 5 [8 marks]

z-transforms

An LTI system has transfer function

$$H(z) = \frac{3 + z^{-1}}{1 + 1.5z^{-1} - z^{-2}}.$$

a. Find the poles and zeros of $H(z)$. **[1]**

b. List all possible ROCs. For each ROC, find $h(n)$ using partial fractions. **[3]**

c. Which ROC gives a stable system? Is that stable system causal, anticausal or noncausal? Explain. **[2]**

d. Assuming the system is causal, write the difference equation that describes it. **[1]**

e. The stable system from part (c) is driven by $x(n) = \cos(\pi n)$ for all n. Find the output $y(n)$. **[1]**

Question 6 [7 marks]

Digital filters

a. A data-acquisition system samples at $F_s = 1$ kHz and picks up a 125 Hz interference tone. Design a second-order FIR notch filter

$$H(z) = b_0 \left(1 - 2\cos\omega_0\, z^{-1} + z^{-2}\right)$$

that removes 125 Hz completely and has $|H(\omega)| = 1$ at $\omega = \pi$. Give ω_0 and b_0. **[2]**

b. Find the gain $|H(\omega)|$ of your notch filter at 0 Hz and at 250 Hz. Comment briefly on what this says about the notch width. **[1]**

c. Does this notch filter have a stable causal inverse? Explain using its zeros. **[1]**

d. A different signal is sampled at $F_s = 1.2$ kHz and is corrupted by a 200 Hz hum and all of its harmonics. You decide to use the moving-average comb filter

$$y(n) = \frac{1}{M+1}\sum_{k=0}^{M} x(n-k).$$

Find the value of M that places nulls at 200 Hz and its harmonics. List every frequency in $0 < F \le F_s/2$ that is removed. **[2]**

e. What is the delay (in samples and in ms) that this comb filter introduces? What is its DC gain? **[1]**

Question 7 [9 marks]

IIR filter design

A digital lowpass filter is required to meet these specifications:

- Passband ripple: ≤ 3 dB
- Stopband attenuation: ≥ 15 dB
- Passband edge frequency: 1 kHz
- Stopband edge frequency: 2 kHz

- Sampling frequency: 8 kHz

The filter is to be designed by applying the bilinear transformation to an analogue Butterworth filter. Take $T = 1$.

a. Find the digital band edge frequencies ω_p and ω_s. **[1]**

b. Prewarp them to obtain the analogue edge frequencies Ω_p and Ω_s. **[1]**

c. Convert the specifications to linear scale and find the Butterworth order N. **[2]**

d. Find the cutoff frequency Ω_c so that the stopband specification is met exactly, and write the analogue transfer function $H(s)$. **[2]**

e. Assuming the answer to part (d) is $\Omega_c = 0.85$ rad/s, apply the bilinear transformation to obtain $H(z)$ with $a_0 = 1$. Write the difference equation of the digital filter, and check H at $\omega = 0$ and $\omega = \pi$. **[3]**

Question 8 [6 marks]

FIR filter design

A causal linear-phase FIR filter of length $M = 7$ is to approximate the ideal lowpass response

$$H_d(\omega) = \begin{cases} 1, & |\omega| < \pi/4 \\ 0, & \pi/4 \le |\omega| \le \pi. \end{cases}$$

a. Find the impulse response $h_d(n)$ of the ideal filter. **[1]**

b. Obtain the FIR impulse response $h(n)$, $0 \le n \le 6$, by the windowing method with a **Hamming** window. Give the values to 4 decimal places. **[3]**

c. Is $h(n)$ symmetric or antisymmetric? State the group delay of the filter. **[1]**

d. Compute $H(0)$. Explain why it is not 1, and say how you would make the DC gain exactly 1. **[1]**

End of questions